Cairo University – Faculty of Engineering
Petrochemical Engineering Program
Credit-hour System



# STYRENE MONOMER PRODUCTION

## *Mobil–Badger EBMax Process*

## *A Case-Study*

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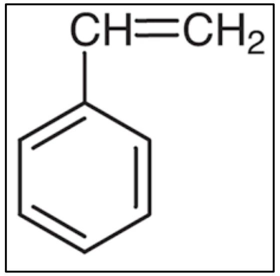
E. STYRENICS
Egyptian Styrenic & Polystyrenic Production Company

# About Styrene

Styrene is an organic compound with the chemical formula $C_6H_5CH{=}CH_2$. This derivative of benzene is a colorless oily liquid with BP of 145$^{o}$C, although aged samples can appear yellowish. The compound evaporates easily and has a sweet smell, although high concentrations have a less pleasant odor.

| | |
|---|---|
| **Molecular Formula** | **$C_6H_5CH{=}CH_2$** |
| Empirical Formula (Hill Notation) | $C_8H_8$ |
| Molecular Weight | 104.15 g/mol |
| Density | 0.909 g/cm$^3$ at 20°C |
| Color | Colorless |




<u>Uses of Styrene:</u>

Styrene is primarily used in manufacturing synthetic rubbers, latex, polystyrene resins which are further employed in making disposable tableware, insulations, pipes, plastic packaging, automobile and electronics part, printing cartridges, containers, etc.

## Global Styrene Demand

| % | STYRENE APPLICATION | METRIC TONNES |
|---|---|---|
| 56.2% | Polystyrene (solid, foam, and film) | 16,944,000 |
| 22.2% | Styrene Copolymers (ABS, SAN, etc.) | 6,678,000 |
| 6.1% | SB Latex | 1,851,000 |
| 4.8% | SB Rubber | 1,445,000 |
| 3.4% | Composites | 1,035,000 |
| 7.2% | Other | 2,183,000 |
| 100.0% | TOTAL | 30,136,000 |

Source: IHS Markit, 2020 Edition: Spring 2020 Update, Annual Supply and Demand Balances

# E-STYRENICS

## The Egyptian Styrene and Polystyrene Production Company

E-styrenics is planned to produce 200,000 tons of polystyrene based on 8000 hours per year.

E-styrenics able to cover the demand of high quality polystyrene – produced by its plant- to all companies using polystyrene in Egypt. The Polystyrene project is planned to be a two Phases project.

The 1st phase is to produce polystyrene in our plant out of exported styrene components.

The 2nd phase is to construct a styrene production plant out of local Benzene and Ethylene components.

The Polystyrene Process facility uses the INEOS/Lummus technology for the manufacture of High Impact Polystyrene (HIPS) and General Purpose Polystyrene (GPPS) as well as certain ISBL support facilities.

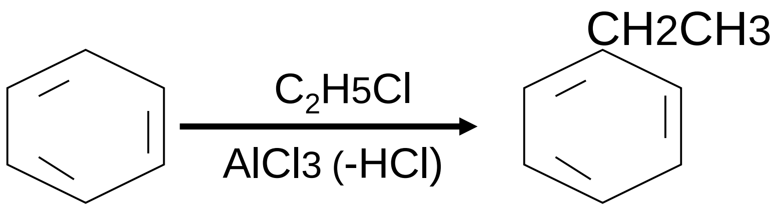
The Egyptian company started commercial production at its 200,000 tons/year plant in Alexandria, Egypt in 2013.
The company's shareholders included the Egyptian Petrochemicals Holding Company (ECHEM) as well as the Egyptian National Investment Bank and the Ministry of Finance.

In May 2019, the Egyptian Ministry of Petroleum has decided to close the local PS producer E-Styrenics' plant permanently due to mounting debts that have reached millions of Egyptian pounds.

# Styrene Production

- The conventional method of producing styrene involves the alkylation of benzene with ethylene to produce ethylbenzene, followed by dehydrogenation of ethylbenzene to styrene.

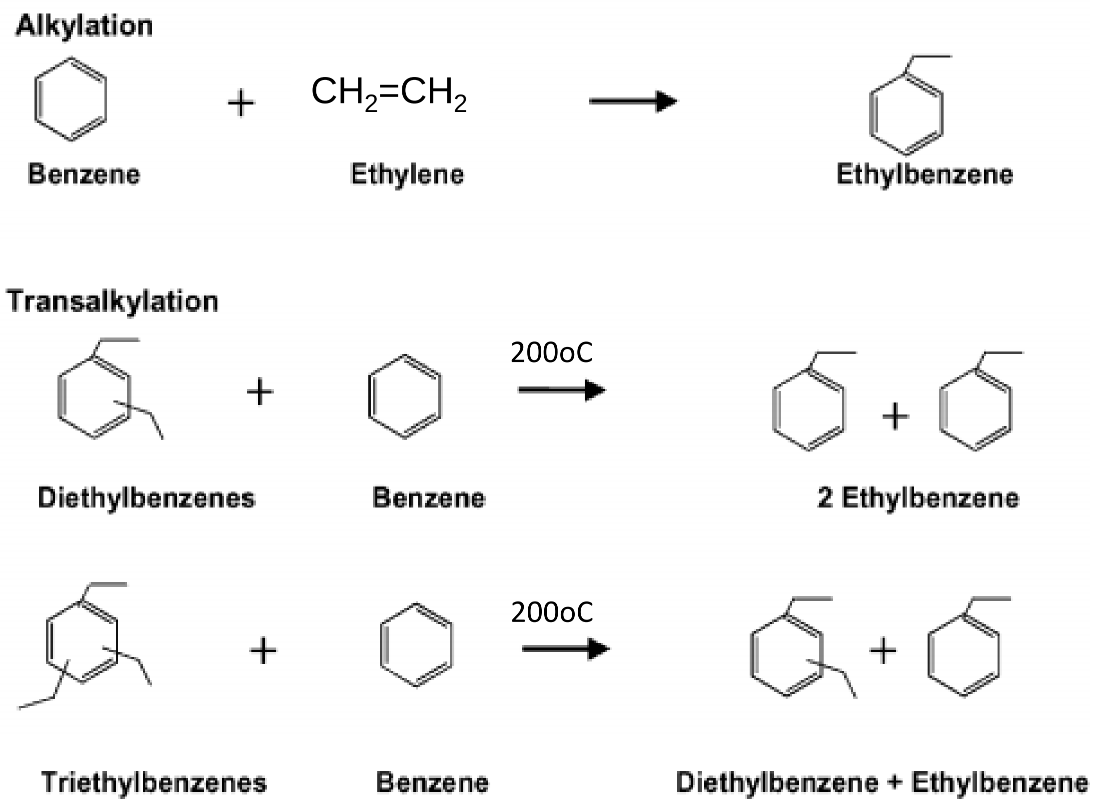
## 1. Alkylation (Friedel Kraft Alkylation)



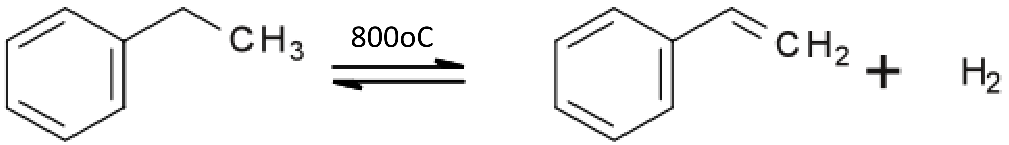

benzene ethyl benzene

- Reaction of ethyl chloride or ethylene with benzene in presence of anhydrous $AlCl_3$ catalyst gives ethyl benzene.
- The reaction is exothermic, forward and yields in addition to EB, polyethylbenzenes as side products.
- The benzene should be dry as presence of water deactivates the catalyst.

## Alkylation & Transalkylation for Ethylbenzene Production

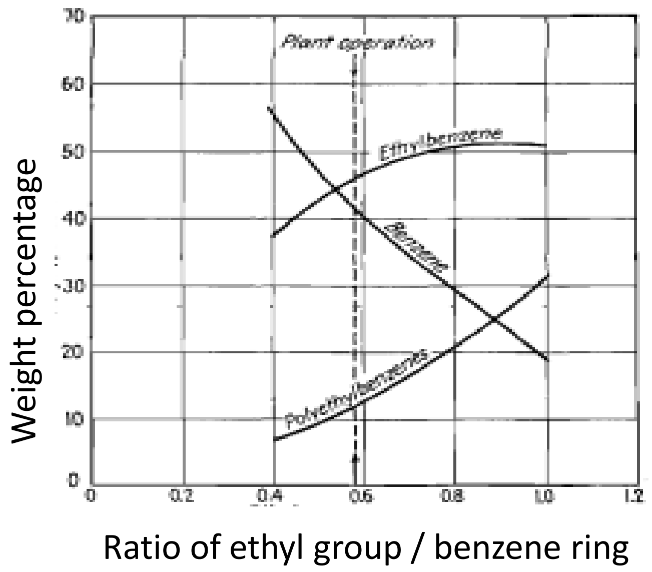



## Dehydrogenation of Ethylbenzene for Styrene Production



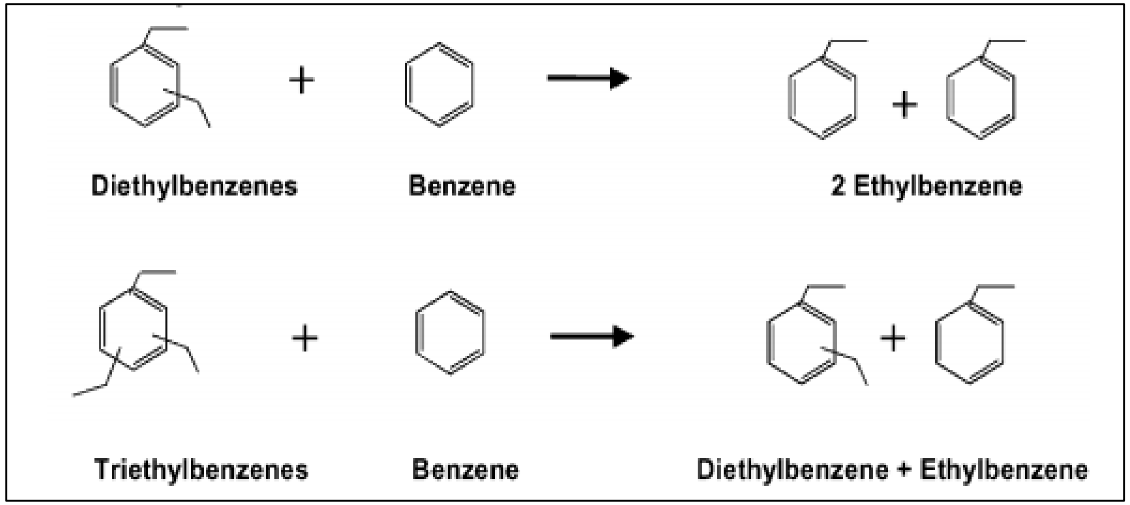
# Styrene Production

- The ratio between the 2 reactants is very important.
- At a specific temperature, the ratio between the B and E should be carefully selected to give the minimum amount of PEB and maximum amount of EB.
- The optimum ratio is 1 B : 0.6 E; i.e. benzene is used in excess of the stoichiometric amount.



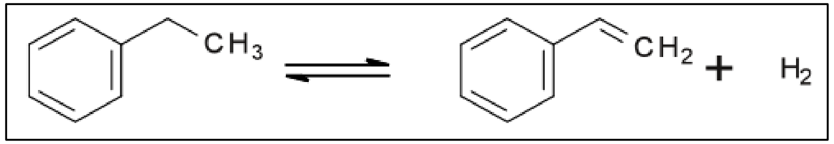

## 2. Transalkylation (Dealkylation of Polyethyl benzenes)

- PEB (about 20%) react with excess benzene at 200$^{o}$C and atmospheric pressure in presence of catalyst to give EB & B.
- The reaction is a liquid phase <u>endothermic reaction.</u>




## 3. Dehydrogenation of Ethylbenzene

The dehydrogenation reaction is reversible, catalytic and <u>endothermic reaction</u>, takes place at 800$^{o}$C.



# Ethylebenzene (EB) Production Overview

- Ethylbenzene is an organic compound with the formula $C_6H_5CH_2CH_3$. It is a highly flammable, colorless liquid with an odor similar to that of gasoline. Its boiling point is 136$^oC$ and density 0.87 g/cm3.

- Nearly 40% of the world's ethylbenzene capacity still utilizes the $AlCl_3$-based Friedel–Crafts alkylation process first introduced in the 1950s. This process operates at low benzene:ethylene (B:E) ratio (3:1) and reasonable temperatures (90◦C).

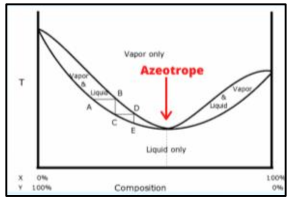
- However, over the past 20 years, the operating costs associated with the corrosivity of $AlCl_3$, and other problems associated with its safe handling and disposal have induced most manufacturers to move toward zeolite catalyzed processes.

- Zeolite catalyzed processes are licensed by Mobil–Badger, Lummus–UOP, CDTech, and Dow Chemical.

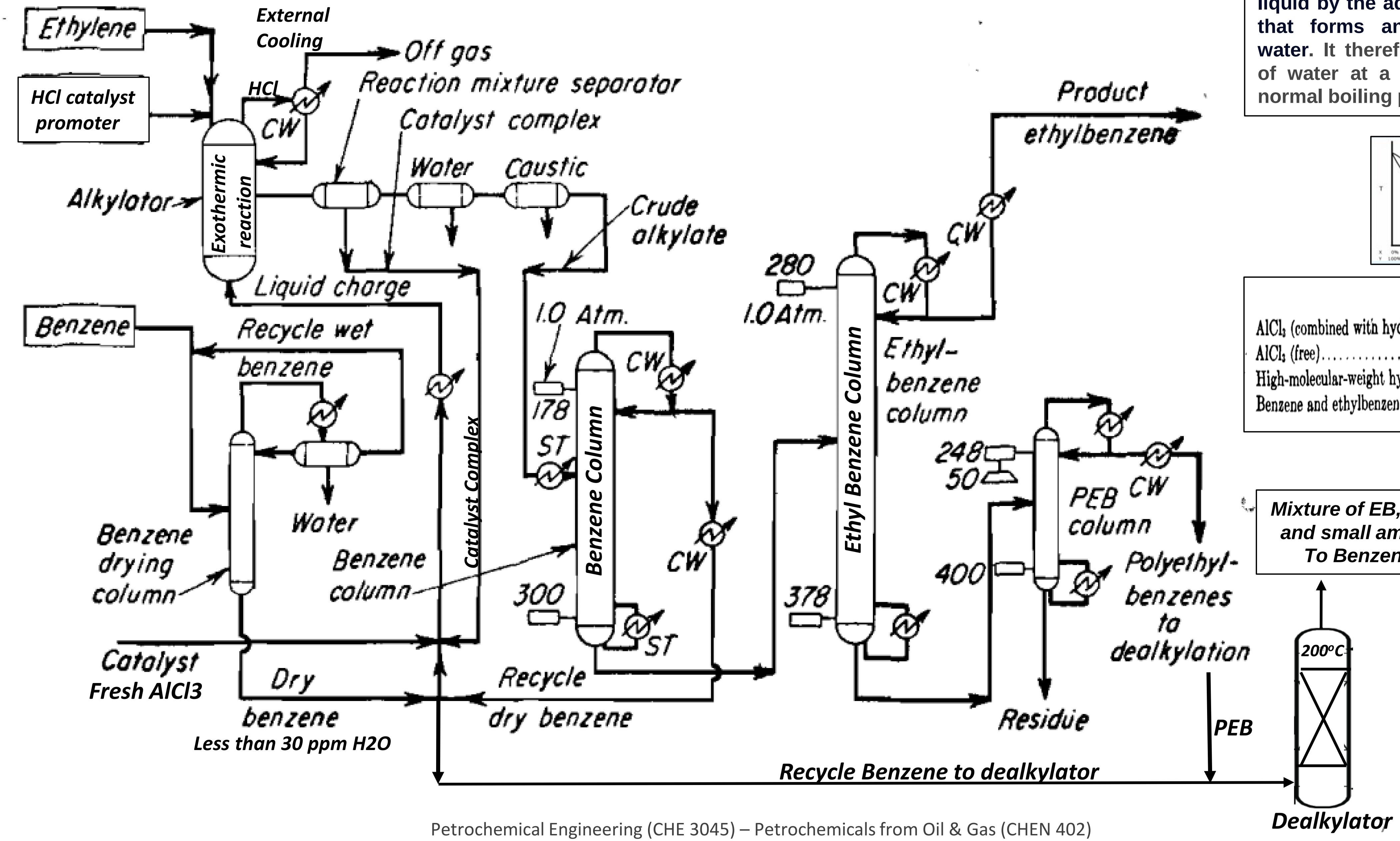
- Over the last several years, the industry has moved to more selective ethylbenzene dehydrogenation catalysts, making the purity of ethylbenzene of even greater importance. The selectivity to the monoethylated product is very important in ethylbenzene synthesis, whereas ethylene oligomerization is normally a secondary consideration.

# EB Production Flow Diagram

**Azeotropic Drying:**
**The process of removing water from a liquid by the addition of another liquid that forms an *azeotrope with the water.** It therefore allows the removal of water at a temperature below the normal boiling point of 100°C.




| | Per cent |
|---|---|
| $AlCl_3$ (combined with hydrocarbons) | 26 |
| $AlCl_3$ (free) | 1 |
| High-molecular-weight hydrocarbons | 25 |
| Benzene and ethylbenzene | 46 |




Mixture of EB, unreacted B and small amount of PEB To Benzene Column

# EB Production Process Description (1)

1. Gaseous ethylene is passed into the liquid benzene in presence of the catalyst and HCl catalyst promoter at temperatures 40 – 100°C.

2. Benzene of purity 99% is pumped through an azeotropic drying column to dry the benzene up to less than 30 ppm water.

3. The dry benzene is mixed with the recirculated catalyst complex and fresh $AlCl_3$ catalyst and is then sent to the alkylator.

4. In the alkylator, B is converted to EB and PEB at 95°C and pressure slightly above atmospheric.

5. As the reaction is exothermic, external cooling takes place at the top of the alkylator, where off gas (mainly HCl) is boiled off, cooled and returned back to the alkylator.

6. After the alkylator, the first separator separates the crude alkylate (B, EB and PEB) from the catalyst complex.

7. In the second separator washing with water and then settling where wash water gets out.

# EB Production Process Description (2)

8. In the third separator neutralization of the crude alkylate with NaOH solution takes place.

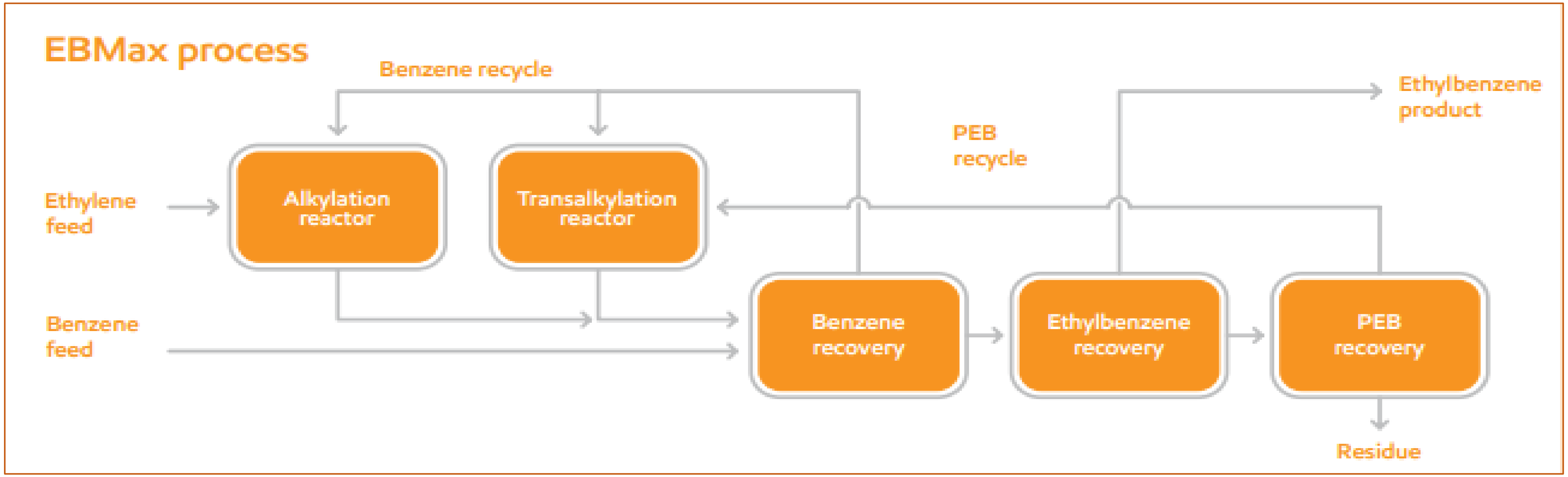
9. The crude alkylate, containing about 40% B, 20% PEB and the rest being EB, is fed to a train of conventional distillation columns where the first and second operate at atmospheric pressure to yield dry benzene and EB, respectively.

10. The B obtained at the top of the first column is sent to the B drying column together with fresh feed B before being fed to the alkylator.

11. The bottom stream from the first column being EB and PEB is fed to the second column where product EB is taken from the top and PEB together with small amounts of heavy residue are separated at the bottom and sent to the third column.

12. The third column operates under vacuum at 50 mmHg (WHY?) and separates PEB at the top to deliver it to the dealkylation unit (transalkylation unit) together with B from the BC. Product from the top of the dealkylator (mixture of EB, unreacted B and small amount of PEB) is fed to the BC together with the product of the alkylator.

# EB Production - Mobil–Badger EBMax Process (Overview)




**Alkylation Reactor**

An alkylation reactor uses ExxonMobil catalyst to convert benzene and ethylene to ethylbenzene (EB) in the liquid phase. A small fraction of the EB is further alkylated to polyethylbenzenes (PEB), which is recovered in distillation and converted back to EB in the transalkylation reactor.

**Transalkylation Reactor**

A transalkylation reactor uses ExxonMobil catalyst to convert the small amount of PEB with benzene in the liquid phase to EB. The effluent is sent to distillation to recover the additional EB production.

**Purification**

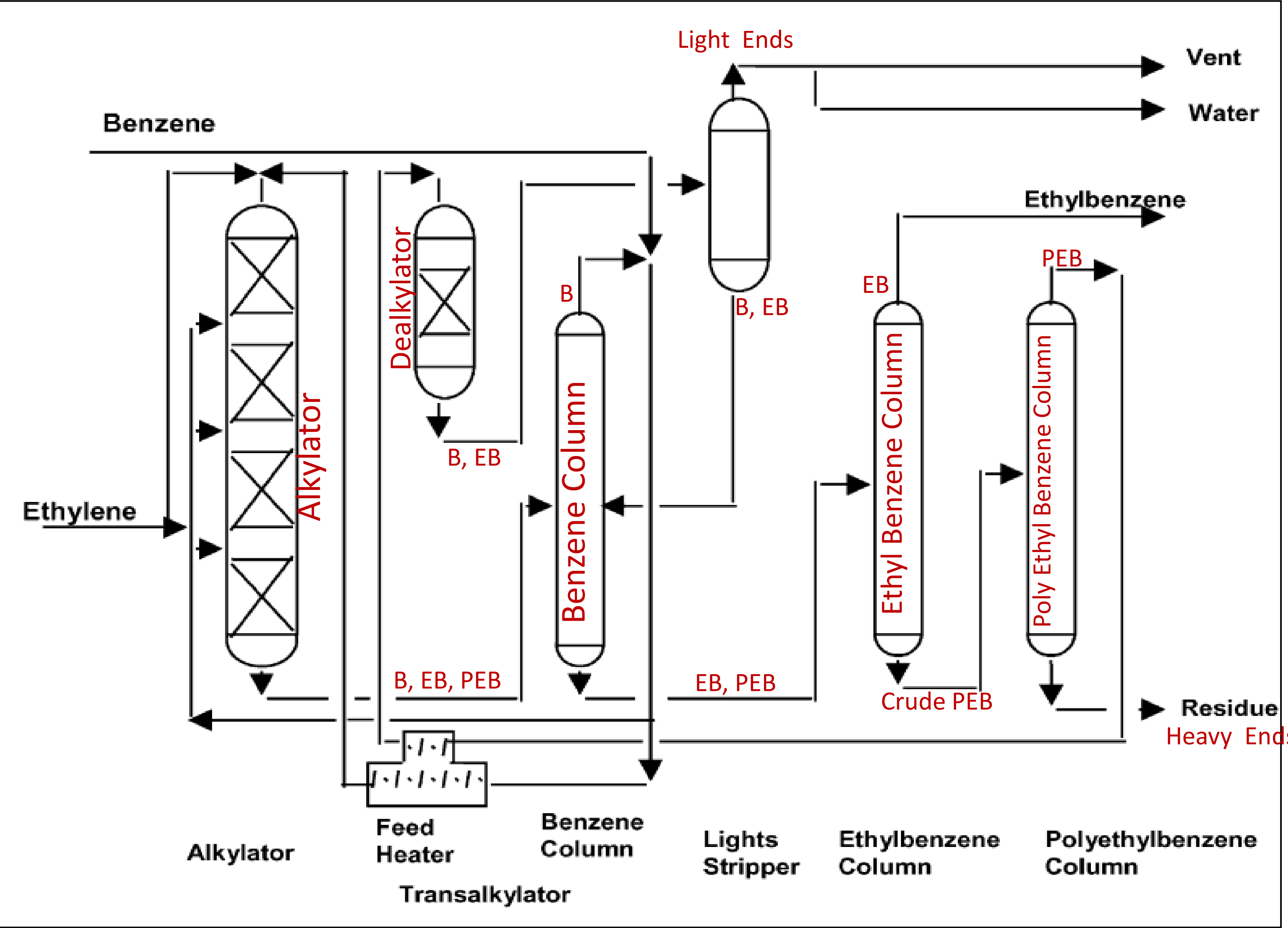
A simple distillation train recovers unreacted benzene, produces EB product, and recycles PEB to transalkylation.

# Mobil–Badger EBMax Process

- The EBMax process consists of an alkylation section, a transalkylation section and a distillation section.

1. **Alkylation Section:**

Ethylene is reacted with benzene in a liquid filled reactor equipped with multiple fixed zeolite catalyst beds. A portion of the total ethylene feed to the plant is injected upstream of each bed. The benzene feed is heated to reaction temperature prior to being introduced to the first reactor bed. The reaction heat is used to generate steam.



# Mobil–Badger EBMax Process

**2. <u>Distillation Section:</u>**

It consists of <u>three columns, each of which generates steam in its overhead system (How? Explain)</u>. The benzene recovery column recovers unreacted benzene from the reactor effluent. Distillate from the benzene recovery columns is recycled to the reactor system, while its bottoms feed the second distillation column (EB column) where the ethylbenzene product is recovered as a liquid distillate. The bottoms product from the second column feeds the PEB recovery column, which recovers recyclable PEB for processing in the transalkylation reactor. Bottom product (Heavy Ends) from the PEB recovery column, is withdrawn from the process as a residue stream and used as a fuel.

**3. <u>Transalkylation:</u>**

Polyethylbenzenes are reacted with benzene to form ethylbenzene. <u>*Polyethylebenzenes recovered in the distillation section are mixed with benzene, heated to reaction temperature and fed to the transalkylation reactor.*</u> The secondary reactor contains a single bed of Mobil's TRANS 4 transalkylation catalyst. The secondary reactor effluent is combined with the alkylator effluent and fed to the benzene recovery column. <u>Nearly all the reaction heat and energy input to the process is recovered (Explain)</u> through steam generation. Rejection of heat to cooling water occurs only in vent condensers and coolers on streams flowing to offsite tankage.

EB purity can be up to 99.9 %, depending on the extent of benzene removal designed into the benzene recovery column.

# Mobil–Badger EBMax Process Special Features (1)

Zeolites are porous, crystalline aluminosilicate minerals known for their unique adsorption, ion-exchange, and catalytic properties.

## Mobil-Badger EB Max Catalyst

1. EBMax is a liquid phase ethylbenzene process that uses Mobil's very stable catalyst zeolite which is more monoalkylate selective than large pore zeolites and thus allows the process to use low feed ratios of B : E.

2. The lower B : E ratios reduces the benzene circulation rate which, in turn, improves the efficiency and reduces the throughput of the benzene recovery column.

3. Because the process operates with a reduced benzene circulation rate, plant capacity can be improved without adding distillation capacity. This is an important consideration, since distillation column capacity is a bottleneck in most ethylbenzene process units.

# Mobil–Badger EBMax Process Special Features (2)

**Vapor Phase versus Liquid Phase Transalkylation**

1. Polyethylbenzenes (PEB), almost exclusively diethylbenzenes, undergo transalkylation with recycled benzene in a second reactor (transalkylator). Mobil–Badger offers both liquid phase and vapor phase transalkylation processes.

2. The vapor phase process is very efficient in removing benzene feed co-boilers such as cyclohexane, methylcyclopentane, propyl and butylbenzenes.

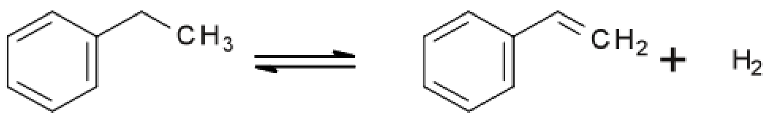
3. Because the EBMax process produces very low levels of propyl and butylbenzenes, the more energy efficient liquid phase process is preferred.

# Mobil–Badger EBMax Process Special Features (3)

**Mobil-Badger EB Max Operating Conditions**

1. The EBMax process operates at low temperatures, and therefore the level of xylenes in the EB product is very low, typically less than 10 ppm.
2. The process operates with low benzene to ethylene ratio which reduces the capital investment and consumption of HP steam.
3. The temperatures ranging from 370 to 420◦C and pressures ranging from 0.69 to 2.76 MPa (6.8–27.2 bar).
4. The energy efficiency of the Mobil–Badger process is reported to be high as all of the exothermic heat of reaction and energy input to the process is recovered through steam generation.

# Dehydrogenation of EB to Styrene



- The reaction is an endothermic reversible catalytic vapor phase reaction which takes place as 600 – 800ºC.
- Steam minimizes coking and also reacts with formed coke to give $CO_2$.
- The output from the dehydrogenator contains almost 50% styrene, 45% unreacted EB and the rest being aromatics (B & T), tar, $H_2$, $H_2O$.

**Reaction Conditions:**

Temp: 600 – 800ºC
Press.: 200mmHg (slightly less than atmospheric pressure)
Catalyst: $Fe_2O_3$ with $K_2O$ promoter
Supplementary promoters: oxides of Cr, Sn, Pt and Ti – prevents deposition of coke on the active sites of the catalyst.
Steam : EB = 7 : 1 (mole %) or more.

**Pressure:**

The super heated steam (900ºC) in the ratio up to 15:1 steam : EB acts as a diluent which shifts the reversible reaction forward, resulting in decreased partial pressure of EB and thus increases the conversion of EB to Styrene (40% per pass), the total being 91%.

**Function of Steam:**

1. Lowers the partial pressure of the reactants & shifts reaction forward.
2. Minimizes coking (coke blocks the pores of the catalyst).
3. Reacts with formed coke to give $CO_2$.
4. Controls side reactions thus selectivity reaches 70%.
5. Provides part of the heat for the endothermic dehydrogenation.

# Factor Affecting Selectivity & Conversion

1. **Pressure**
2. **Steam**
3. **Catalyst**
4. **Temperature**
5. **Space velocity**

- Decrease in pressure enhances conversion (Why?)
- High steam to EB ratio increases conversion (Why?)
- Catalyst raises the selectivity and conversion.
- Space velocity = Volumetric flow rate of reactants / volume of catalyst ($time^{-1}$)
- There is a proportional relationship between Temp. & Space Velocity.

**Space Velocity & Temperature**

- Space velocity affects the temperature needed to achieve the required conversion.
- The inlet temp. must be increased as space velocity is increased and vice versa.
- As the catalyst losses activity due to loss of potassium promoter, the temp. should be raised to compensate the loss of catalyst activity and thus maintain constant conversion.
- When the reactor temp. reaches the mechanical design limit, the space velocity must be reduced to extend the catalyst life.
- As the reaction is endothermic, if no heat is added the temperature will drop and the conversion in turn will not reach 40%. To achieve an economical conversion the process is carried out in at least two stages with intermediate heating.

# Mobil–Badger Styrene Process Catalyst

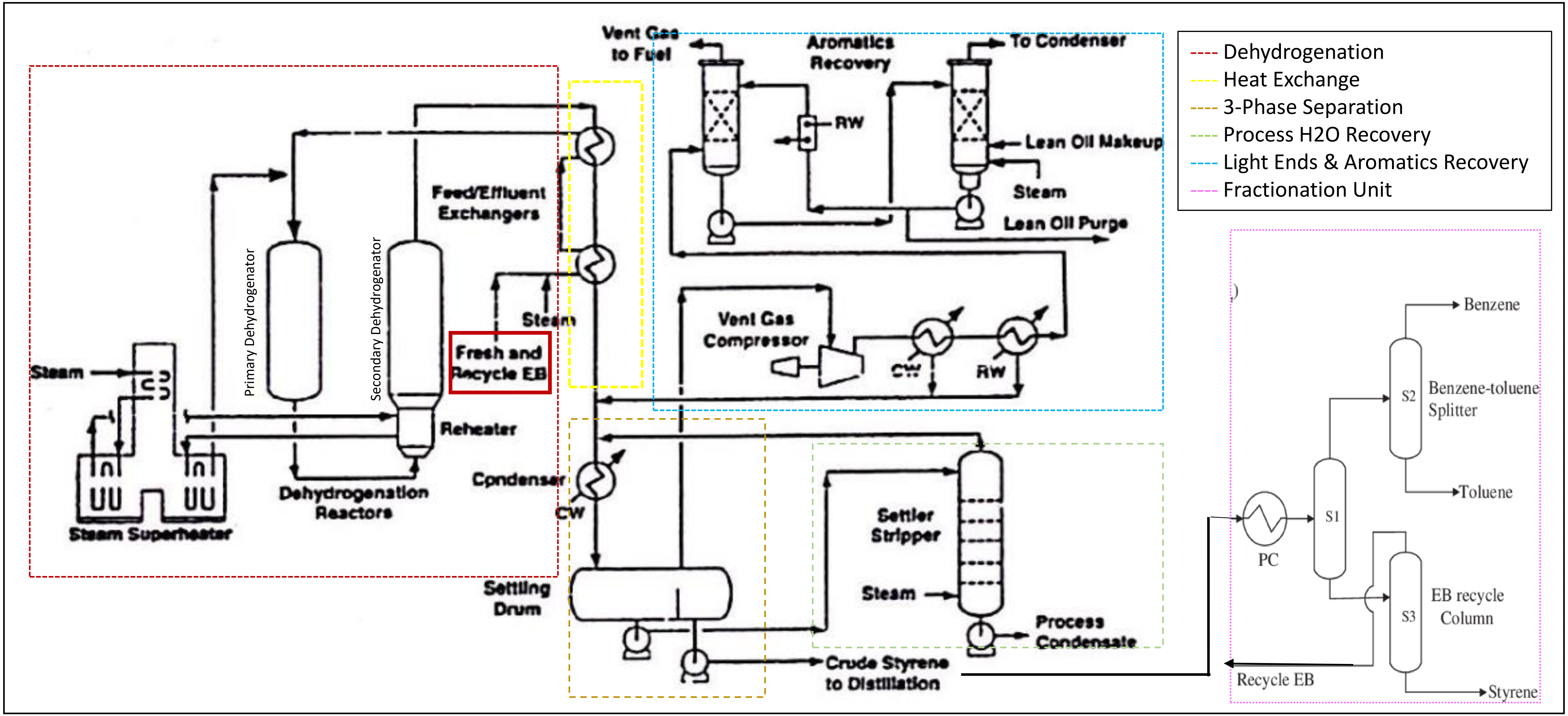
**Mobil-Badger Catalyst:**

- The catalyst consists of ferric oxide with potassium oxide as promoter to increase the selectivity.
- Catalyst: $Fe_2O_3$ with $K_2O$ promoter
- Supplementary promoters: oxides of Cr, Sn, Pt and Ti – prevents deposition of coke on the active sites of the catalyst.
- The catalyst losses its activity due to loss of potassium promoter.

**Catalyst Stabilization Technology (CST):**

- Badger developed a technology to modify the dehydrogenation catalyst which involves injecting $K_2O$ (volatile activator) while the plant is in operation.
- Consequently, the catalyst activity loss is greatly slowed down and thus the catalyst lifetime improved by nearly 50%.

# Dehydrogenation of EB to Styrene Process Flow Diagram



# Dehydrogenation of EB to Styrene Process Description (1)

1. The main types of equipment in the dehydrogenation section of the plant are the 1. steam superheater, 2. the primary and secondary dehydrogenation reactors, and 3. a series of feed/effluent exchangers. High pressure steam is also generated by the recovery of heat from the reactor effluent stream.

2. Fresh and recycle EB are mixed with low pressure steam and then enters the effluent HE where the feed is vaporized in the first and then superheated in the second and mixed with additional steam. Before entering the primary dehydrogenator, steam from the super heater is thoroughly mixed with EB-steam feed to ensure homogeneous feed mixture. The effluent from the primary reactor is reheated by exchange with superheated steam in a HE located directly before the secondary reactor.

3. The dehydrogenation reactors are designed to provide low pressure drop and uniform flow distribution. The inlet temperature sets the EB conversion to 40%. The reactor effluent is cooled in a series of three heat exchangers that heat the EB and steam feed to the reactors and generate steam.

4. The effluent from the dehydrogenation reactors comprises styrene, unreacted EB, steam, $CO_x$ and a variety of by-products, including benzene and toluene as well as heavier & lighter compounds.

# Dehydrogenation of EB to Styrene
# Process Description (2)

5. On leaving the dehydrogenation section, the effluent is cooled against the reactor feed in effluent/feed HEs. Then the cooled reactor effluent is condensed in the crude styrene condenser and then sent to a settling drum where aqueous phase ($H_2O$), liquid HC (styrene-containing aromatic phase) and vent gas are separated.

6. The aqueous phase containing traces of dissolved HCs is directed to a stripping column where HCs and light gases (mainly $CO_2$) are removed by steam stripping. The condensate is stripped and used as boiler feed water for steam generation.

7. The carbon dioxide generated by the decoking reaction dissolves in and separates with the aqueous phase making it acidic, which could cause corrosion in downstream equipment as well as hinder the separation of the organic and aqueous phases. Thus, a <u>corrosion inhibitor is added upstream of the condensation step</u> to reduce or eliminate the acidity of the aqueous phase.

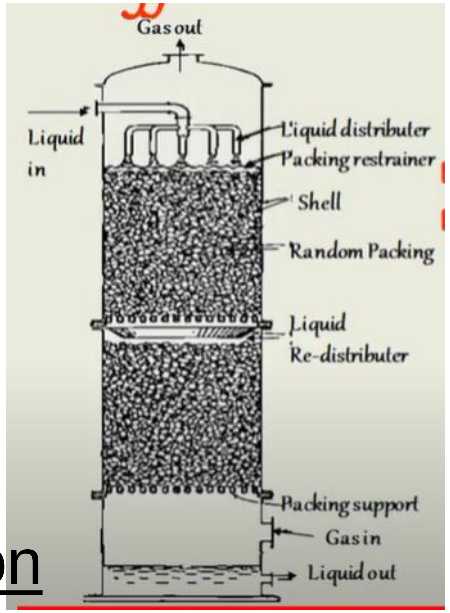
# Dehydrogenation of EB to Styrene
# Process Description (3)

8. The vent gas consisting of $H_2$, aromatics and methane is sent to a compressor, then cooled in a series of HEs to condense more water and aromatics. The remaining gas is then sent to absorption/stripping system to remove the remaining aromatics from the vent gas. The vent gas, the majority of which is hydrogen, is used as fuel gas.

9. The HC portion (crude styrene) of the reactor effluent is fed to the distillation section of the plant, which consists of three <u>distillation columns operating under vacuum to minimize temperature and inhibit polymer formation</u>.

10. In the first column, benzene-toluene mixture is separated from EB and styrene. The bottoms stream containing EB and styrene is sent to EB recycle column (the second column) where EB is separated overhead from styrene and recycled to the reactor.

# Dehydrogenation of EB to Styrene
# Process Description (4)




Packed Distillation Column

11. <u>The boiling points of EB and styrene are close to each other at 100mmHg (74$^o$C for EB and 82$^o$C for styrene) which makes separation difficult. To accomplish this separation, packed low pressure distillation column is used. This leads to low bottoms temperatures (less than 100$^o$C) which ensures no polymer formation and low tar formation.</u>

12. The bottom stream of the second column is fed to the third column (styrene finishing column) where purified styrene is recovered overhead and heavier compounds mainly polymer and inhibitor as bottoms stream.

13. <u>Styrene is stored in refrigerated inhibited storage tanks to prevent polymer formation during storage.</u>

- Continuous contacting (differential contacting) equipment.
- Basic characteristics: Simple construction; low pressure drop; packings easily made of corrosion腐蚀 resistant materials.
- Applications: (1)Distillation with low pressure drop and <u>vacuum distillation</u>; (2)Absorption and distillation with falling liquid flow rate far greater than upward vapor flow rate; (3)Corrosive materials; (4)Small diameter column; (5)Easily foaming materials; (6)Heat sensitive materials.

# Dehydrogenation of EB to Styrene Process Special Features

- Minimum thermal non-catalytic reactions.
- Very efficient heat conservation.
- Relatively low catalyst volume and prolonged catalyst life (2 – 3 years).
- Reactor design achieves low pressure drop and uniform flow distribution.
- Two vertical reactors in series with intermediate heating ensure enhanced conversion.
- <u>Polymerization inhibitor, vacuum packed distillation towers operating in refrigerated conditions (maximum 10$^{o}$C) prevent styrene from polymerizing into PS.</u>